Answer **all** the questions in this section.

1 Indirect calorimetry is used to estimate energy expenditure in humans by measuring the difference in oxygen concentration between inhaled and exhaled air. During aerobic respiration at 37 ºC, glucose, $C_6H_{12}O_6$, is oxidised to carbon dioxide and water while producing energy as shown in reaction 1 below.

reaction 1 $C_6H_{12}O_6(s) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(l)$ $\Delta H^{\ominus} = -2818 \text{ kJ mol}^{-1}$

(a) **(i)** It is assumed that there are negligible intermolecular forces of attraction between gas particles in an ideal gas.

State **two** other basic assumptions of kinetic theory as applied to an ideal gas. [2]

(ii) Air contains about 21% oxygen, by volume. Exhaled air contains 14% oxygen. At complete rest, a typical adult exchanges approximately 500 cm^3 of air per breath at a rate of 12 times per minute at a temperature of 37 ºC at 1 atmospheric pressure.

By assuming oxygen to be an ideal gas, calculate the volume of oxygen gas consumed per minute. Hence, determine amount of oxygen gas used per minute. [3]

(iii) Assuming that the oxygen inhaled is used for respiration directly, use relevant information in reaction 1 and your answer in **(a)(ii)**, calculate the approximate amount of energy released per minute. [1]

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(b) Glucose, $C_6H_{12}O_6$, is the ubiquitous source of energy for cells in the body.

(i) Define the standard enthalpy change of formation of glucose, $C_6H_{12}O_6(s)$. [1]

(ii) Use the following data, reaction 1 and appropriate data from the *Data Booklet*, to construct an energy cycle and calculate the standard enthalpy change of atomisation of C(s). Show your working.

enthalpy change of formation of $C_6H_{12}O_6(s)$ = –1270 kJ mol^{-1}
enthalpy change of combustion of $H_2(g)$ = –286 kJ mol^{-1}

[3]

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(c) Glucose exists in two forms, α-glucose and β-glucose as shown in Fig. 1.1. If a solution of α-glucose is left some time, it will come into dynamic equilibrium with β-glucose.

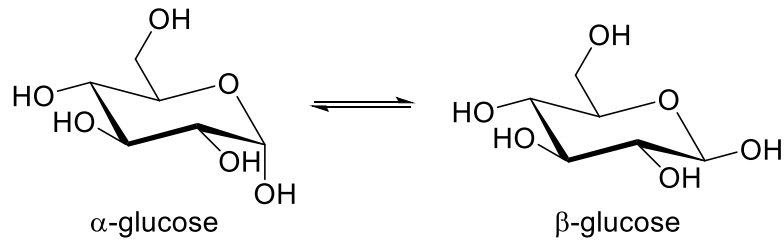

Fig. 1.1

When plane-polarised light is passed through an aqueous solution of glucose, the angle of rotation of the light is dependent upon the structure of the molecule. The angles of rotations of plane-polarised light caused by the two forms of glucose solutions under identical conditions are shown in Table 1.1.

Table 1.1

solution	angle of rotation of plane-polarised light
1.0 mol dm^{-3} of α-glucose	+111°
1.0 mol dm^{-3} of β-glucose	+19°

(i) When 1 dm^3 of a freshly prepared solution of 1.0 mol dm^{-3} α-glucose is left till equilibrium is achieved, the measured rotation is +53°. Assuming that the angle of rotation due to each glucose is directly proportional to its concentration, calculate a value for the equilibrium constant, K_c, for the conversion of α-glucose into β-glucose. [2]

(ii) The conversion of α-glucose into β-glucose is catalysed by acids. State and explain the effect on the final measured rotation if the conversion is now carried out in the presence of dilute sulfuric acid. [2]

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(d) Oxygen used in respiration binds to haemoglobin in red blood cells, which can undergo ligand exchange with water in tissues to regulate oxygen delivery. Both deoxyhaemoglobin and oxyhaemoglobin contain iron atoms in the +2 oxidation state. The oxygen-containing ligand is H_2O in deoxyhaemoglobin, Hb, and O_2 in oxyhaemoglobin, $Hb(O_2)_4$.

One molecule of deoxyhaemoglobin, Hb, can bind with four molecules of oxygen, as shown in equilibrium 1.

equilibrium 1 $\quad Hb(aq) + 4O_2(aq) \rightleftharpoons Hb(O_2)_4(aq) \quad \Delta H^{\ominus} = -630.9\ kJ\ mol^{-1}$

$\Delta G^{\ominus} = -160.7\ kJ\ mol^{-1}$

(i) State *Le Chatelier's Principle*. [1]

(ii) During the initial stage of vigorous exercise, rapid muscle contractions generate heat that spreads through the body, raising core temperature.

Using Le Chatelier's Principle, explain how the initial increase in temperature affects equilibrium 1. [1]

(iii) Suggest how the magnitude of K_c for equilibrium 1 is likely to be and explain its significance. [2]

(iv) Carbon monoxide, CO, can bind to haemoglobin at the same binding site as oxygen. Explain why CO is poisonous. [2]

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